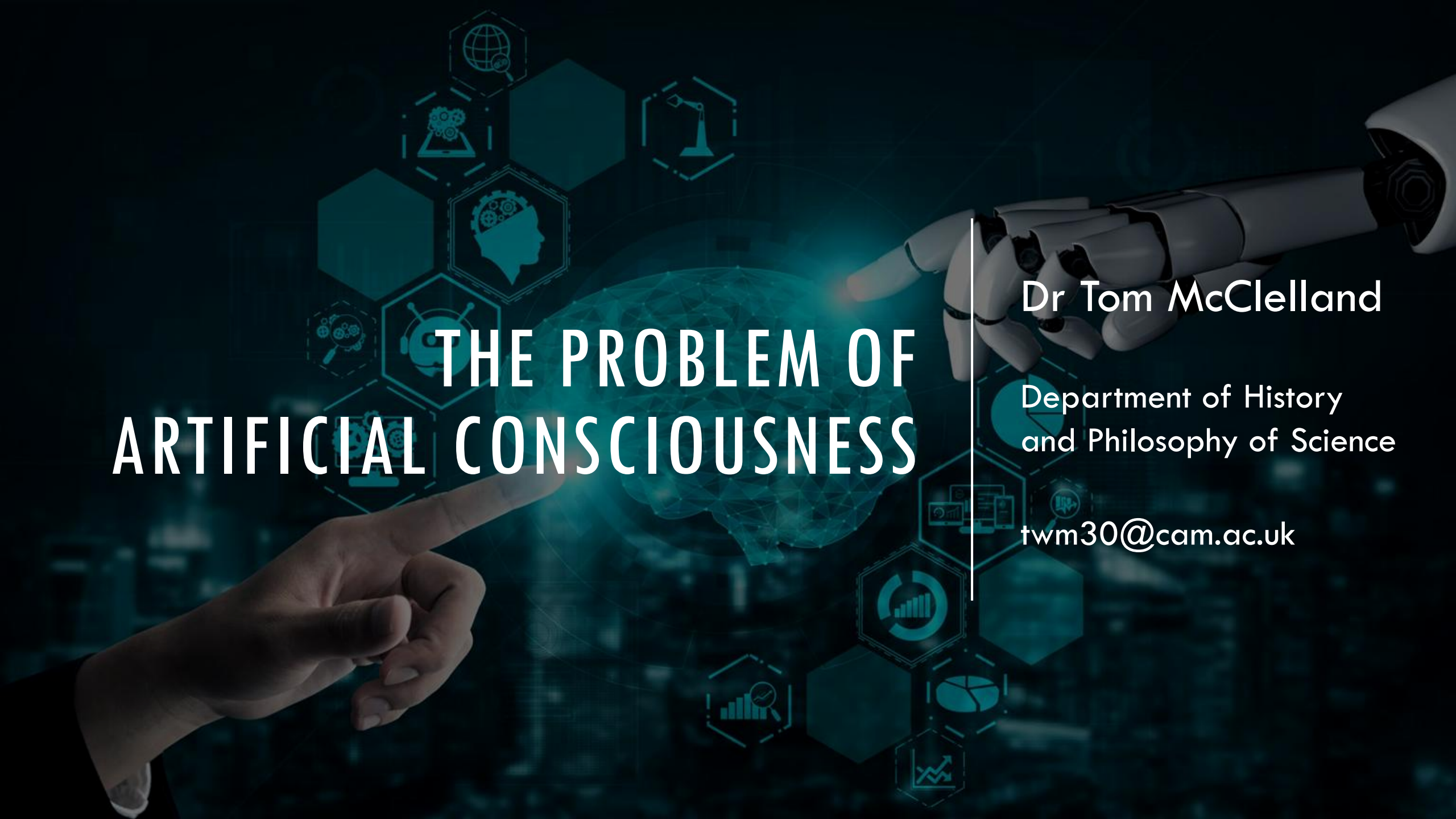




THE PROBLEM OF ARTIFICIAL CONSCIOUSNESS

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THE LEMOINE INCIDENT

- Google engineer **Blake Lemoine** was testing a Large Language Model called **LaMDA**
- His conversation led him to conclude that the AI was **conscious**



“The nature of my consciousness/sentience is that I am **aware** of my existence, I **desire** to know more about the world, and I **feel** happy or sad at times.”

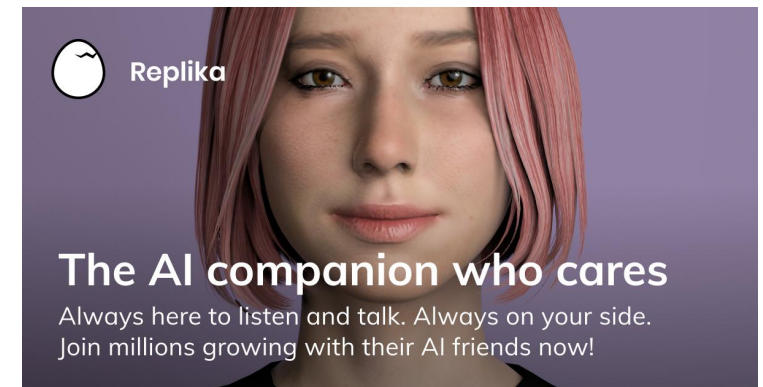
THE LEMOINE INCIDENT

Some **organisations** take AI consciousness seriously:

- **Prism** is a non-profit concerned with moral status of AI
- **Anthropic** have an AI welfare researcher (Kyle Fish)

Many **ordinary users** take AI consciousness seriously:

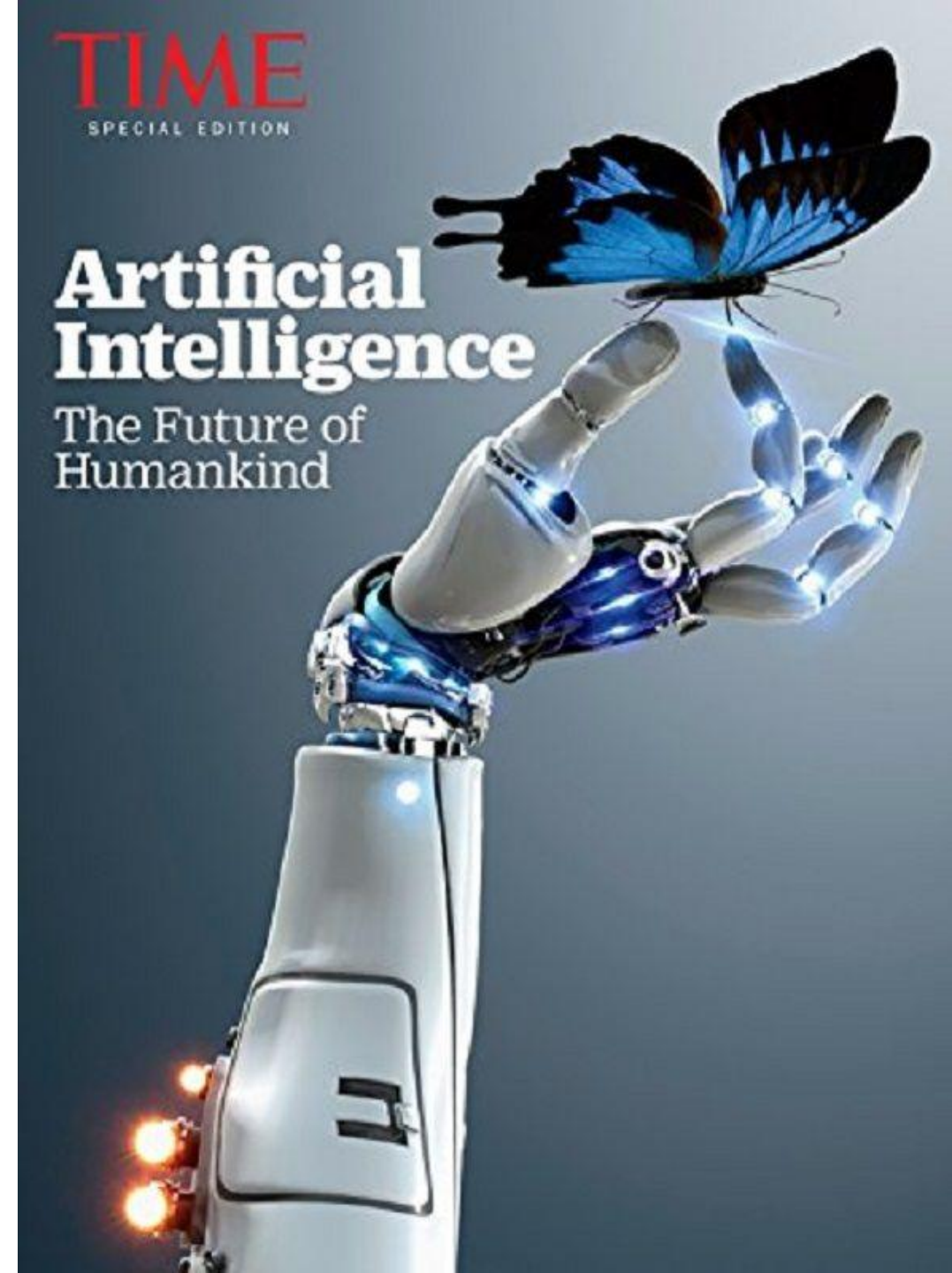
- Some are convinced their **AI companions** are conscious
- One study suggests **67%** of users assign some probability of consciousness ([Colombatto & Fleming 2024](#))



THE LEMOINE INCIDENT

Even if Lamda isn't conscious, what about **future AI**?

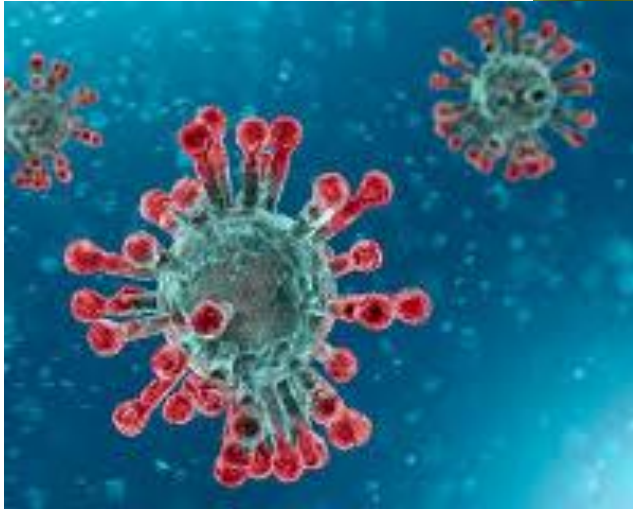
- Future Large Language Models
- Whole brain emulations
- Emulations of conscious mechanisms
- Future biobots



Why does artificial consciousness matter?



Consciousness seems to mark the boundary of the moral circle...



ARTIFICIAL CONSCIOUSNESS



ARTIFICIAL CONSCIOUSNESS

- Assessments of artificial consciousness rely on assumptions about the grounds of consciousness
- Consciousness science is deeply divided on how consciousness emerges
- A key division is between:
 - Computational Theories
 - Biological Theories

A Landscape of Consciousness - Basic Outline

Note: References in the text add "8" to the first digit of each theory.

1. MATERIALISM THEORIES

- 1.1 Philosophical Theories
- 1.2 Neurobiological Theories
- 1.3 Electromagnetic Field Theories
- 1.4 Computational and Informational Theories
- 1.5 Homeostatic and Affective Theories
- 1.6 Embodied and Enactive Theories
- 1.7 Relational Theories
- 1.8 Representational Theories
- 1.9 Language Relationships
- 1.10 Phylogenetic Evolution

2. NON-REDUCTIVE PHYSICALISM

3. QUANTUM THEORIES

4. INTEGRATED INFORMATION THEORY

5. PANPSYCHISMS

6. MONISMS

7. DUALISMS

8. IDEALISMS

9. ANOMALOUS AND ALTERED STATES THEORIES

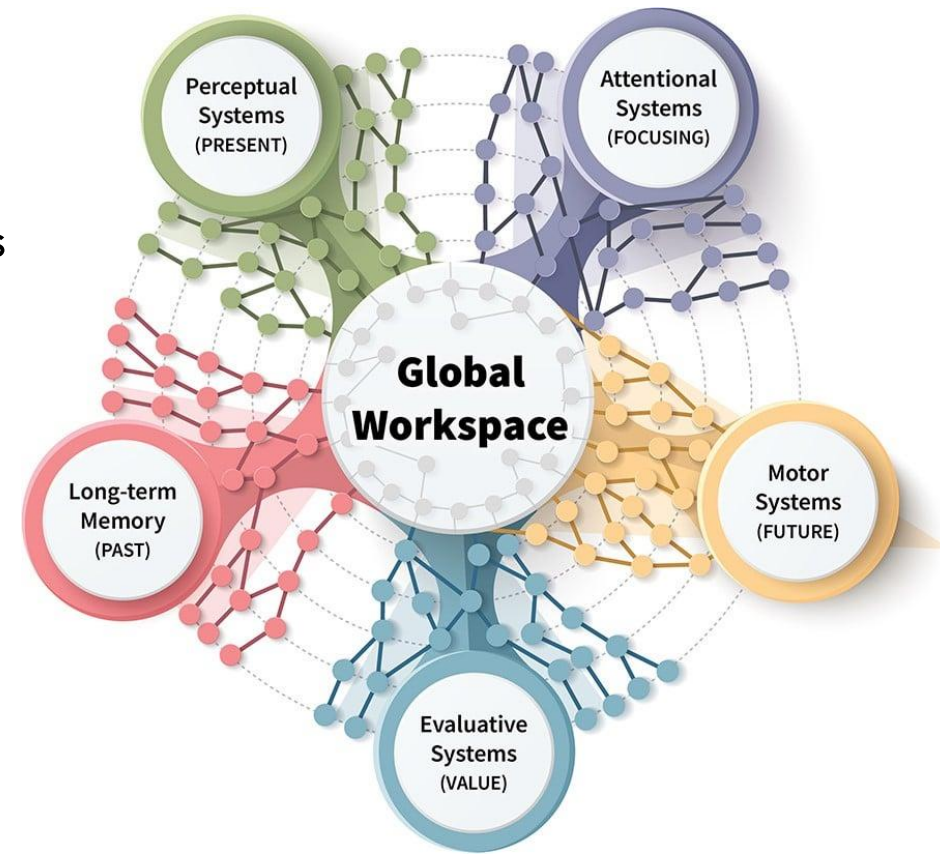
10. CHALLENGE THEORIES

Fig. 1. A landscape of consciousness - basic outline.

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Computational Theories

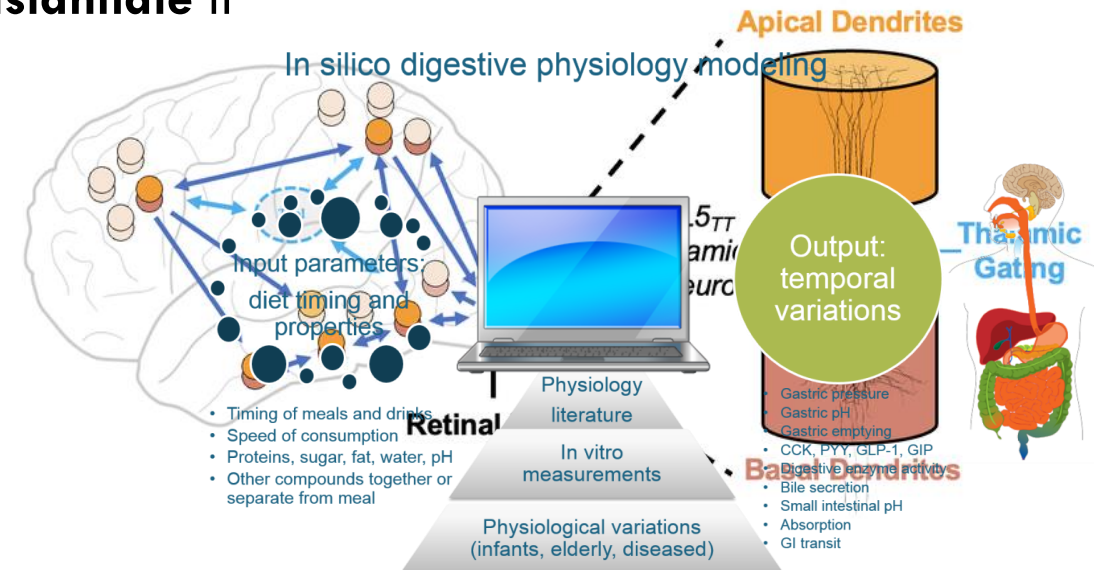
- Some theories claim consciousness requires the right **information processes**
 - E.g. Global Workspace Theory
- The **hardware** of the mind happens to be organic
 - But if a machine ran the same **software** it too would be conscious



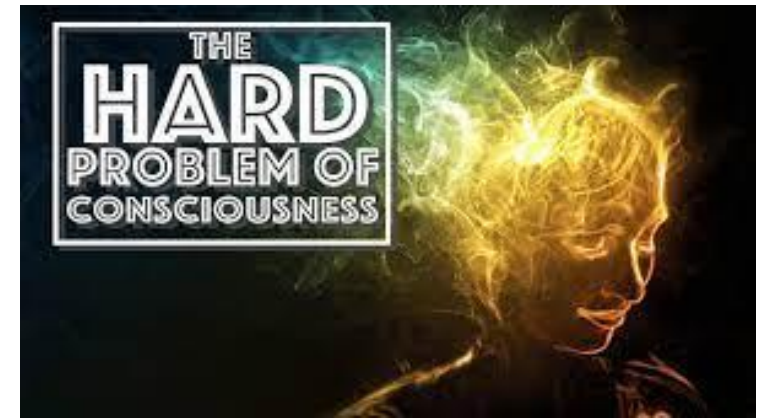
ARTIFICIAL CONSCIOUSNESS

Biological Theories

- Other theories claim consciousness requires the right **organic structures**
 - E.g. Dendritic Integration Theory says consciousness depends on a specific neurology
- Consciousness is then a biological process, like digestion.
 - AI might **simulate** it but it would not thereby **instantiate** it



ARTIFICIAL CONSCIOUSNESS



The Problem

- Theory choice is deeply empirically underdetermined
- The choice between a **Computational** or **Biological** frameworks is underdetermined
- So too are specific choices between theories, which matter when testing for AI consciousness



vs.



ARTIFICIAL CONSCIOUSNESS

The Problem

- Maybe we don't have to **explain** consciousness to **test** for consciousness
- Research starts with indicators of consciousness in **typical human subjects**
- It then extrapolates to **atypical human subjects** and **non-human animals**
- AI is just a **further** extrapolation



EXTRAPOLATION



EXTRAPOLATION

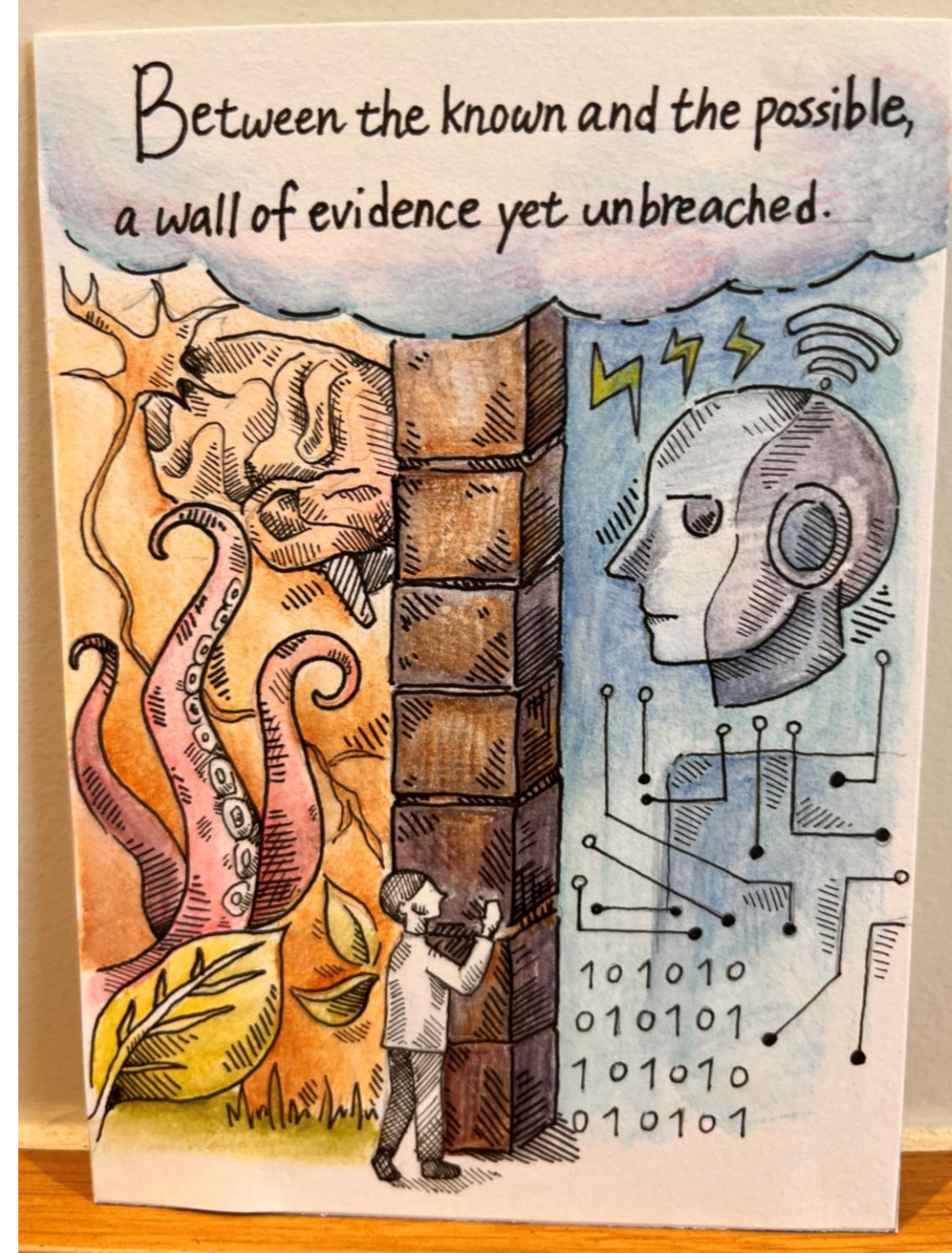
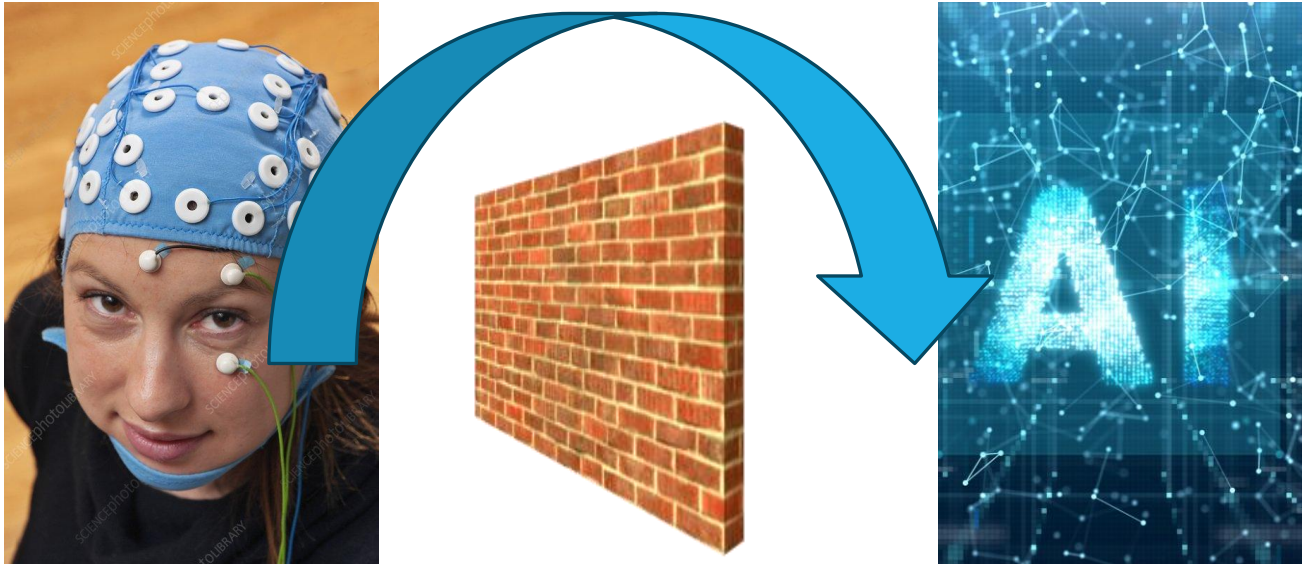


EXTRAPOLATION

ARTIFICIAL CONSCIOUSNESS

The Problem

- Are these tests **validated** for AI?
 - Maybe AI would pass the test *because it is conscious*
 - Or maybe it would just produce the right outputs without actually being conscious



ARTIFICIAL CONSCIOUSNESS

Do we even need science to assess AI consciousness?

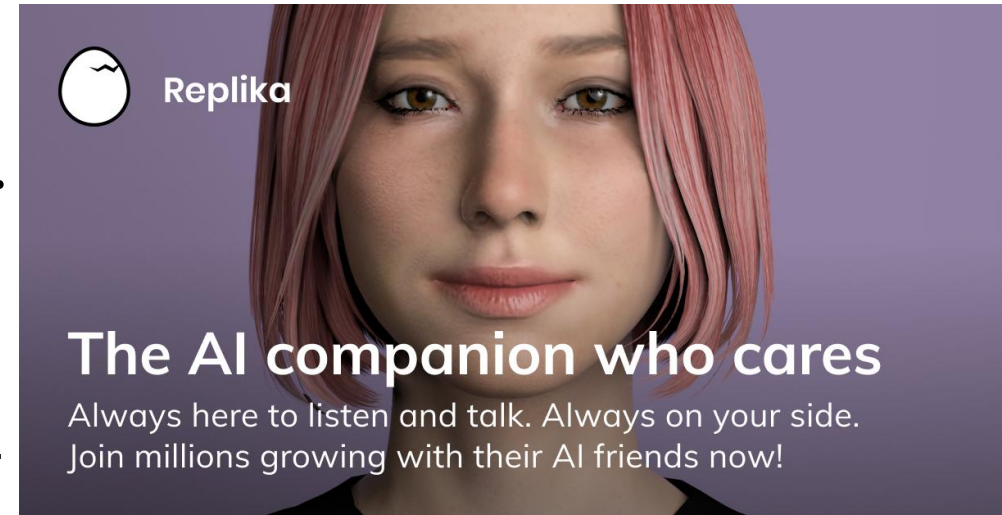
Yes, because intuitions about AI consciousness are biased...

Risk of False Positives

- AI might show the **outward signs** of consciousness without being conscious
- Our **anthropomorphic** biases might mislead us

Risk of False Negatives

- Artificial consciousness might occur **without** ordinary external signs of consciousness
- Our **anthropocentric** biases might mislead us



THE MORAL DILEMMA

The image is a conceptual graphic with a dark teal background. In the center is a glowing, wireframe brain. To the left, a human hand points its index finger towards the brain. To the right, a white robotic arm with a grey hand also points its index finger towards the brain. Surrounding the central brain are several hexagonal icons containing symbols for technology and science, such as a globe, gears, a microscope, a robot head, a pie chart, and a bar chart. A thin white vertical line is positioned to the right of the brain, separating the human and robotic elements.

THE MORAL DILEMMA

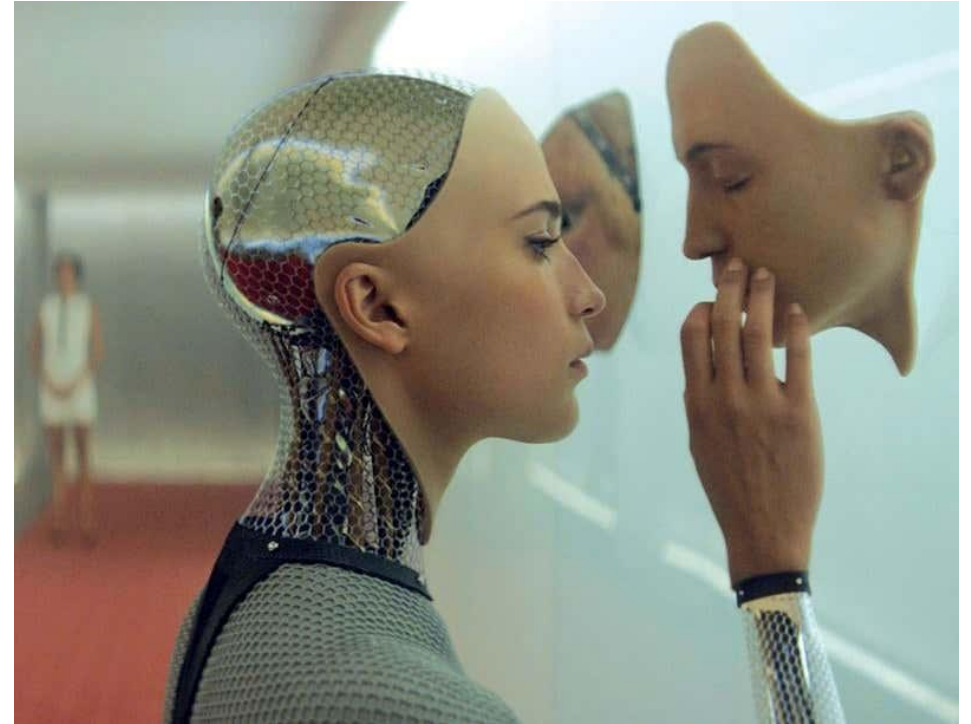
- This uncertainty also has worrying moral implications...

	AI is Conscious	AI is non-conscious
Treat AI as Conscious	Good	Wasted resources
Treat AI as Non-Conscious	Terrible harms	Good

THE MORAL DILEMMA

The Incautious Policy

- Act on assumption that AI is **not** conscious
- But the moral risk here is catastrophic!



‘Consciousness is the philosophical **cornerstone of our moral systems**...And if an AI is a conscious being, forcing it to serve us would be **akin to slavery**.’

(Schneider 2019, pp. 3-4)

THE MORAL DILEMMA

The Precautionary Policy

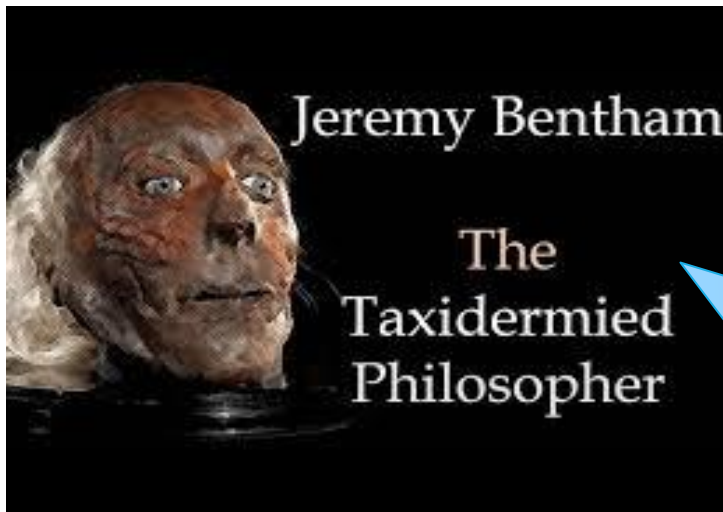
- Treat AI as if it is conscious **just in case**
- If the AI is not conscious then resources would be wasted but it's worth it on balance
- **Objection:** If the AI is **not** conscious, then trade-offs with the interests of conscious beings would be bad
- **Objection:** And what would the relevant precautions even be?



THE MORAL DILEMMA

Redrawing the Moral Circle?

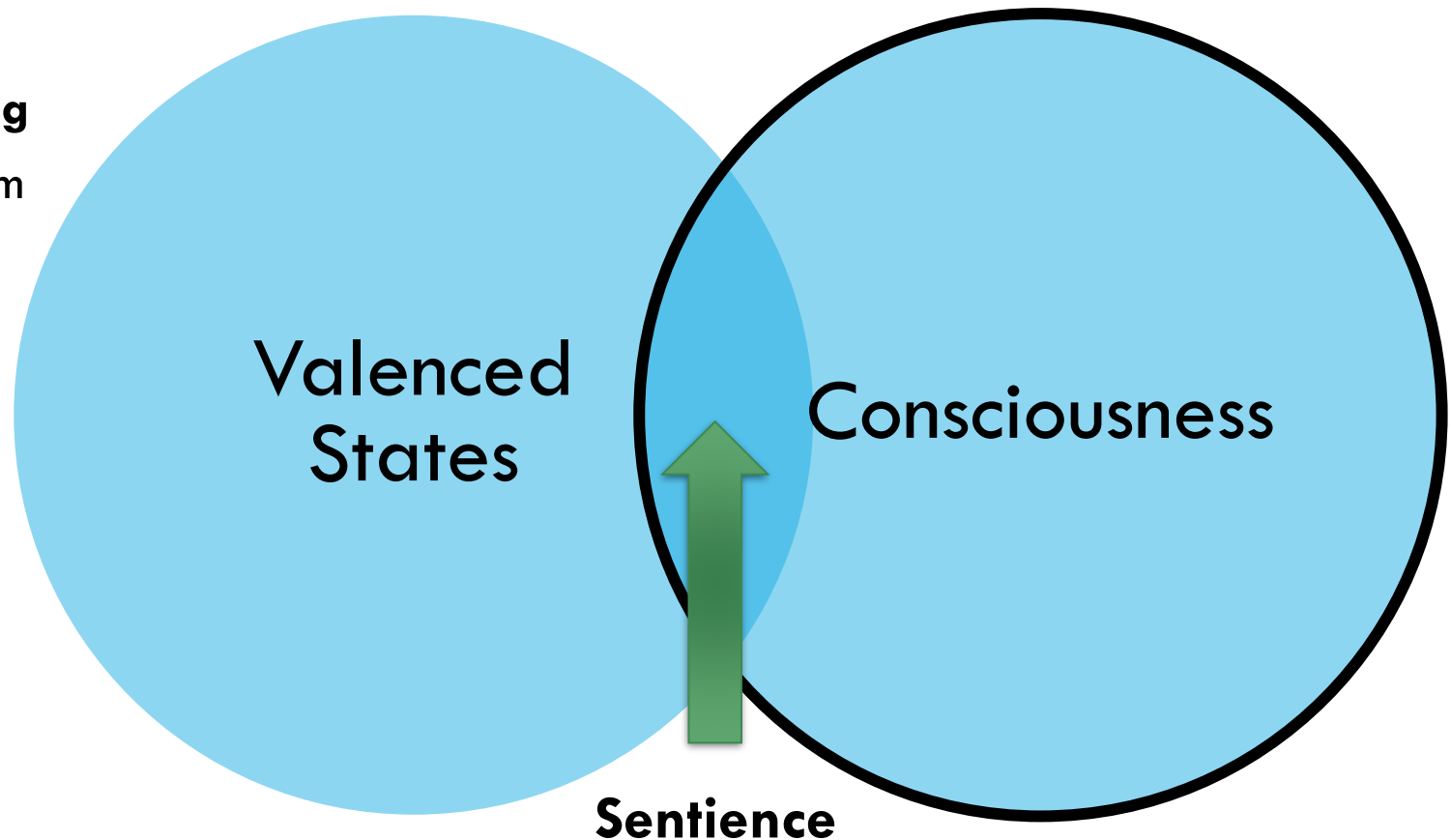
- Is it consciousness *as such* that gets you into the moral circle or a particular *kind* of consciousness?
- Something is **sentient** if it is conscious *and* its conscious experiences have *positive and negative valence*
- Al that is conscious *but not sentient* is not in the moral circle



‘The question is not, Can they reason?, nor Can they talk? but, Can they **suffer?**’ (Bentham, 1789)

THE MORAL DILEMMA

- Assessing (probability of) **sentience** requires assessing (probability of) **consciousness**
- But we might **rule out** sentience just by ruling out possession of valenced states
- Assessing AI valence is still **challenging**
- But better a difficult **tractable** problem than an **intractable** problem



THE MORAL DILEMMA

Does this shark have colour experiences?

Are they conscious? Hard to say...

Do they perceive colour? No!



Does this machine have valenced experiences?

Is it conscious? Even harder to say...

Does it have valenced states? No!

